



"Predictive Modeling of United Kingdom Financial Markets Using Machine Learning: Techniques, Challenges, and Applications from 2020 to 2024"

100701434

University of Derby, Department of Computing and Maths

Abstract

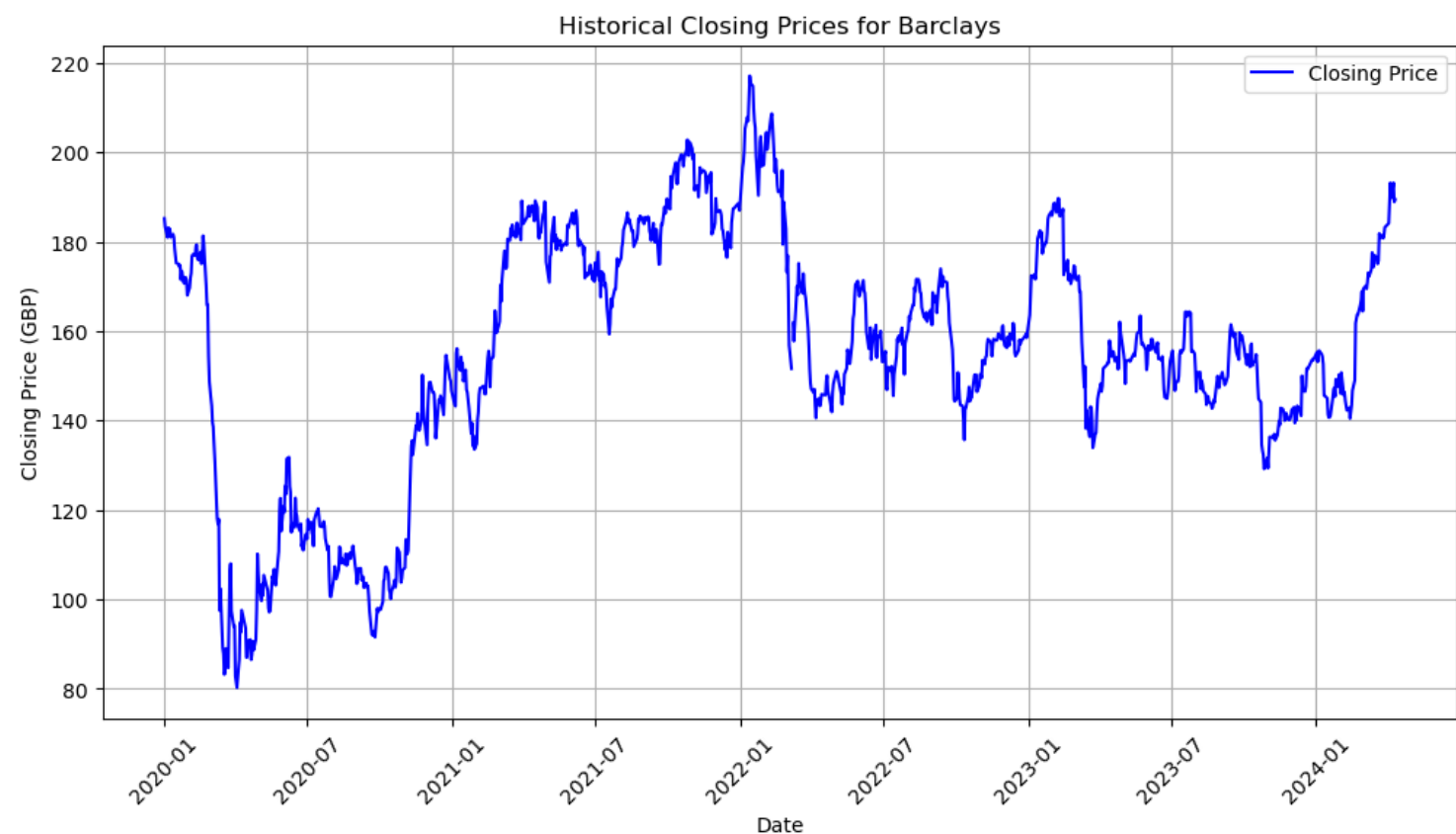
This study delves into the realm of machine learning techniques applied to forecast Barclays stock prices. Employing a combination of exploratory data analysis and feature engineering, we unearthed significant correlations among key features, thereby enriching our predictive capabilities. By scrutinizing the performance of both Linear Regression and Random Forest models, we discovered that the latter exhibited superior accuracy, boasting lower prediction errors. These findings illuminate the efficacy of machine learning methodologies in the domain of financial market prediction, providing valuable insights for informed decision-making amidst the complexities of dynamic financial environments.

Introduction

Financial markets are dynamic and intricate ecosystems influenced by multifaceted factors, including economic indicators, geopolitical events, investor sentiment, and regulatory changes. The ability to accurately forecast market trends and asset prices is important for investors, traders, and policymakers seeking to navigate these complexities. However, traditional econometric models often fall short in capturing the intricate non-linear relationships and temporal dynamics in-built in financial data.

Machine learning (ML) has emerged as a transformative tool in the realm of financial forecasting, offering a data-driven approach that can harness large volumes of historical data to uncover hidden patterns and relationships. Unlike conventional models, ML algorithms excel in handling complex, high-dimensional datasets, enabling them to adapt to changing market conditions and make informed predictions.

This analysis focuses on leveraging machine learning algorithms to predict stock prices for Barclays bank, a leading financial institution, based on historical stock data. By applying ML techniques, we aim to enhance our understanding of market dynamics and improve forecasting accuracy. This academic poster serves as a concise summary of our methodology, key findings, and recommendations derived from this analysis, shedding light on the practical applications of ML in financial market prediction.



Literature Review

The application of machine learning (ML) techniques in financial market prediction has garnered significant attention in recent years due to its potential to extract insights from complex datasets and enhance forecasting accuracy. Researchers have explored various ML algorithms and approaches to address the challenges associated with predicting financial market trends.

Nandi et al. (2023) highlight the complexity of financial markets influenced by economic indicators, political events, and investor sentiment. Their study underscores the utility of ML in analysing financial data, identifying patterns, and making accurate predictions, benefiting applications like stock price forecasting, risk management, and fraud detection. The authors emphasize the importance of understanding both the strengths and limitations of ML techniques for responsible use in financial contexts.

Dhakar and Shiwani (2023) present a survey of contemporary ML techniques used for stock market forecasting. They discuss the challenges posed by the volatility and multi-dimensional nature of stock market data, suggesting that ML offers a robust approach to analyse such complex time series datasets. The authors highlight the experimental results obtained from applying baseline techniques to benchmark datasets, showcasing the comparative effectiveness of different ML techniques in financial market forecasting.

Ryll and Seidens (2019) analysed the effectiveness of various machine learning algorithms in financial market prediction, providing valuable insights into comparative performance and challenges faced in this domain which aligns with my study's focus on predictive modelling in UK financial market using machine learning.

In summary, these studies collectively demonstrate the growing interest in leveraging ML for predictive modelling in financial markets. They underscore the need to address challenges such as data quality, model interpretability, and ethical considerations while harnessing the potential of ML techniques to enhance decision-making and performance in financial forecasting applications.

Methods

In this academic poster a systematic approach encompassing data preprocessing, model selection, and performance evaluation to predict the daily stock price of Barclays bank from 2020 till April 2024 was applied. The dataset was gotten from Yahoo Finance. The dataset containing daily stock price information for Barclays was preprocessed to handle missing values, scale numerical features, and create additional features like moving averages and percentage changes. This step ensured that the data was clean, normalized, and enriched for the analysis.

Two machine learning models were selected for prediction:

- **Linear Regression:** This was chosen to capture linear relationships between input features and the target variable (stock prices).
- **Random Forest Regression:** This was Utilized for its ability to model non-linear relationships and capture complex patterns in the data using ensemble learning techniques.

The performance of the selected models was evaluated using standard metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics provided insights into the accuracy and reliability of the predicted stock prices compared to the actual observed values. This methodological approach allowed for the effective application of machine learning algorithms to forecast stock prices for Barclays, demonstrating the practical utility of predictive modeling in financial markets.

Results/Analysis

The analysis of predictive modeling for Barclays stock prices using machine learning techniques has provided valuable insights into the behavior of financial data and the performance of different modeling approaches.

Exploratory Data Analysis (EDA) revealed significant correlations between key features such as the High, Low, and Adjusted Close prices, indicating important relationships within the dataset. Through feature engineering, additional predictive variables like moving averages and percentage changes were created, enhancing the model's ability to capture relevant patterns.

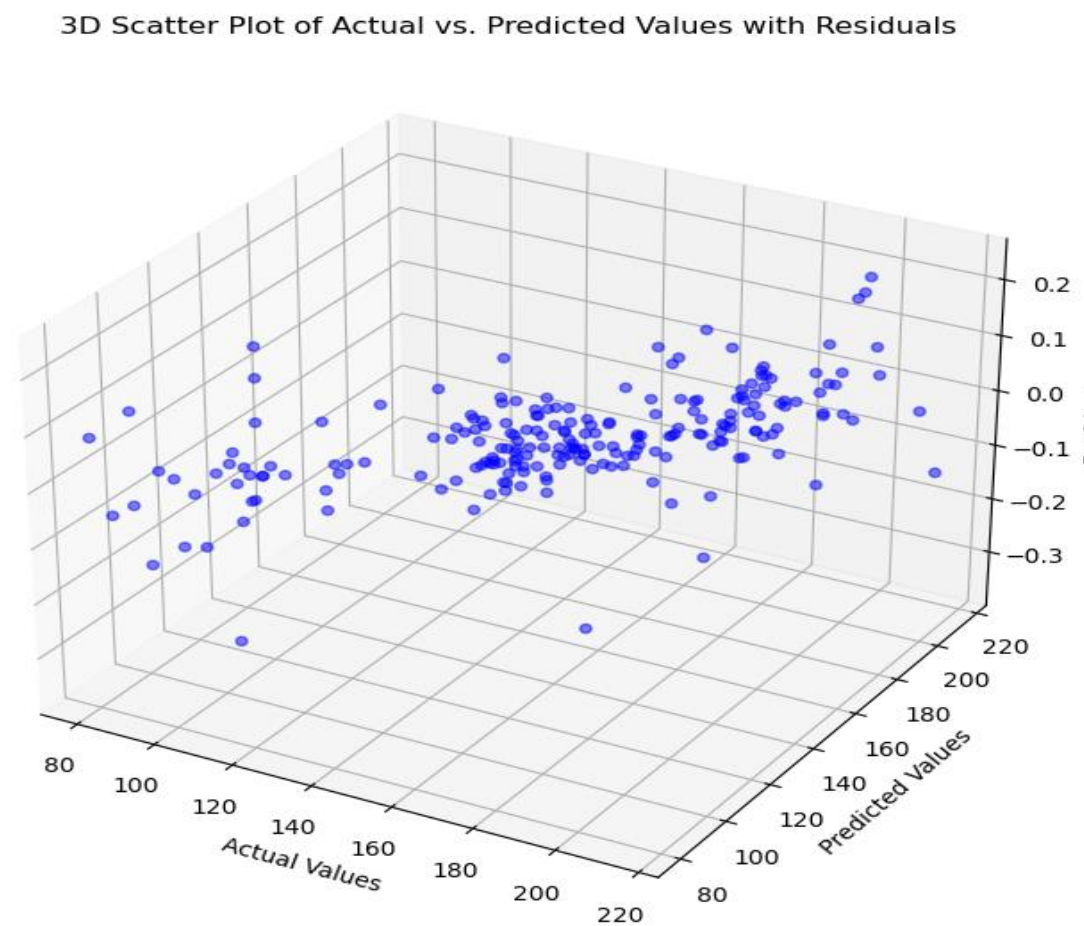
During the data preprocessing stage, missing values were detected in several columns. To address this, missing values were imputed with the mean of the respective columns. Additionally, outliers were detected in the Volume column, which were handled using winsorization to mitigate their impact on the analysis.

Both Linear Regression and Random Forest Regression models were employed to predict stock prices. The evaluation metrics, including Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), demonstrated that the Random Forest model outperformed Linear Regression. The Random Forest model exhibited lower prediction errors of 1.25, indicating its superior performance in stock price forecasting tasks.

The coefficients associated with each feature in the Linear Regression model were analyzed to understand their importance in predicting stock prices. The analysis revealed that High, Low, Open, and Adj Close prices had positive coefficients, indicating a positive correlation with the target variable (stock price), while the Volume had a negligible coefficient.

Residual analysis was conducted to assess the performance of the Linear Regression model. The distribution of residuals showed that, on average, the model's predictions were unbiased and consistent across the range of data. However, there were some outliers with larger residuals, suggesting instances where the model's predictions deviated significantly from the actual values.

The findings underscored the effectiveness of machine learning techniques, particularly Random Forest models, in financial market prediction. The study contributes actionable insights that can be leveraged by investors and analysts to make informed decisions in dynamic financial environments. By leveraging advanced modeling techniques and feature engineering, machine learning enables more accurate and robust predictions of stock prices, enhancing decision-making capabilities in the financial sector.



Conclusion

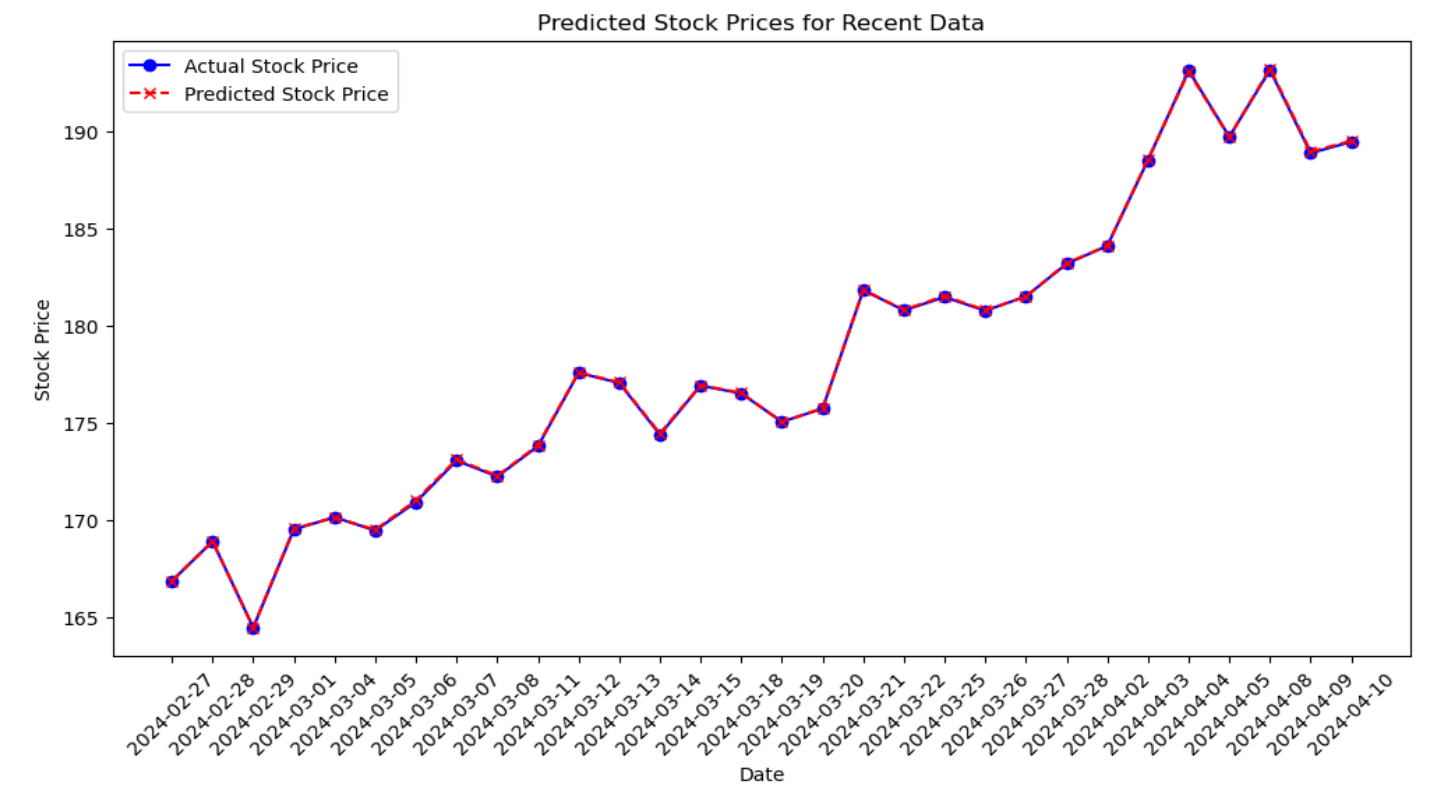
This study on predictive modeling for Barclays stock prices using machine learning techniques has provided valuable insights into financial data analysis and forecasting. By employing exploratory data analysis (EDA), feature engineering, and model evaluation, the research demonstrates the significance of advanced techniques in understanding market behaviors and improving prediction accuracy.

The importance of this study lies in its potential to enhance decision-making for investors and analysts in dynamic financial environments. Accurate stock price predictions enable informed decisions on investment strategies, risk management, and portfolio optimization.

Future applications of this work include deploying developed models in real-time trading platforms and extending methodologies to analyze other financial instruments and markets. Advancements in machine learning algorithms and alternative data sources offer opportunities for further refinement and broader applications.

Key findings highlight the superiority of Random Forest models over traditional Linear Regression approaches in stock price forecasting. While Random Forest models exhibit lower prediction errors, ongoing efforts should focus on refining data preprocessing techniques and enhancing model interpretability for future iterations of projects of this type.

In conclusion, this study contributes to advancing financial data analysis and forecasting by demonstrating the efficacy of machine learning techniques. It lays a foundation for further exploration and innovation in predictive modeling for financial markets.



References

Scan the QR code below to see the references for this project.



Contact

Email Address: 100701434@unimail.derby.ac.uk